

Yuming Feng

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EDUCATION

Department of Electrical Engineering, Stanford University

Stanford, USA

Master of Science, Electrical Engineering (Expected)

Sept. 2025 – Jun. 2027

School of Physics, Peking University (Ranked #1 in China - QS 2025)

Beijing, China

Bachelor of Science, Physics

Sept. 2021 – Jun. 2025

Coursework: Deep Learning, Human-Centered LLMs, Data Structure and Algorithm, Multi-Agent Systems, Natural Language Processing with Deep Learning, Apps with LLMs Inside

PUBLICATIONS

[1]. **Yuming Feng***, Chuye Hong*, Yaru Niu*, Shiqi Liu, Yuxiang Yang, Wenhao Yu, Tingnan Zhang, Jie Tan, and Ding Zhao. **Learning Multi-Agent Collaborative Manipulation for Long-Horizon Quadrupedal Pushing**, Published in *International Conference on Robotics and Automation (ICRA)* 2025. [paper](#) [website](#) [github](#)

[2]. Zili Tang, **Yuming Feng** and Meng Guo. **Collaborative Planar Pushing of Arbitrary Objects with Multiple Robots in Complex Scenes**, Published in *Robotics: Science and Systems (RSS)* 2024. [paper](#) [website](#)

PROFESSIONAL EXPERIENCE

AI Product Builder

Stanford, USA

IdeaWeave(self-employ)

Nov. 2025 - Mar. 2026

- Built end-to-end telemetry and artifact-based debugging for the diagram pipeline, and an auto prompt-engineering pipeline to iterate on prompts systematically without overfitting to single cases.
- Designed and implemented a four-stage LLM pipeline (Planner→Judge→Executor→Validator) that combines ReAct-style tool use (info search) with reflection loops for plan and DSL revision in a single workflow.
- Reduced cost by ~87% and generation latency by ~60% through a compact DSL, model routing, and prompt caching, while preserving diagram quality.
- Defined the core product experience as a full creation loop: intent clarification → diagram generation → canvas refinement → one-click recording, reducing friction between thinking and communication
- Introduced persona-based personalization and persistent user context, enabling more relevant multi-turn interactions; early analytics showed ~1.9 sessions/user and ~4.3 page views/session
- Led go-to-market through targeted creator outreach and launch content, generating 100K+ impressions and 300+ product users in 3 weeks

Machine Learning Infrastructure Intern

Beijing, China

Galbot

Mar. 2025 - May. 2025

- Contributed to develop a high-performance reinforcement learning training platform leveraging JAX and MuJoCo, designed for modular and scalable learning policy across diverse physical intelligences.
- Engineered a distributed training pipeline with optimized data flow and GPU parallel computation strategies, achieving an 83% reduction in training time compared to legacy PyTorch implementations.
- Designed a flexible abstraction layer and API interfaces to support heterogeneous task configurations, enabling rapid experimentation across different simulation setups, and authored comprehensive technical documentation to streamline team collaboration and platform extensibility.

RESEARCH EXPERIENCE

Empirical Study of LLM Alignment and Fine-Tuning Strategies

Stanford, USA

Independent research project under Stanford CS 224N [github](#)

Jan. 2026-Mar. 2026

- Conducted a controlled study of alignment pipeline hand-off, analyzing optimal SFT→DPO transition across paraphrase detection (Quora Question Pairs, 283k examples) and Shakespearean sonnet generation.
- Implemented a GPT-2 (124M) training framework comparing Full Fine-Tuning and LoRA parameterization, showing that FFT outperforms LoRA by ~3% Acc / F1 points while LoRA enables up to 2× larger batch sizes.
- Demonstrated that alignment gains diminish under small-model, low-data regimes, where DPO provides limited improvements over strong SFT baselines and LoRA consistently trails full fine-tuning.

SumForU — LLM Alignment for Persona-Steerable Review Summarization

Stanford, USA

Independent research project under Stanford CS 329X [arXiv](#) [github](#)

Sept. 2025-Dec. 2025

- Built the training framework for persona-steerable summarization, combining SFT (asymmetric distillation) with RLAIF (PPO), achieving an average improvement of ~70% across key metrics over the base model.
- Designed a comprehensive qualitative + quantitative evaluation pipeline spanning rule-based, LLM-based, and user-based metrics, establishing measurable indicators for alignment, grounding, and decision utility in complex summarization tasks.
- Evaluated zero-shot transfer on unseen categories, with the RL-aligned model delivering an average performance gain of ~40% over the base model, demonstrating strong generalization and data efficiency.

NoveltyRank — LLM-based Retrieval-Augmented Novelty Estimation

Stanford, USA

Independent research project under Stanford CS 230 [arXiv](#) [website](#) [github](#)

Sept. 2025-Dec. 2025

- Demonstrated that pairwise comparison task formulation outperforms binary classification for conceptual novelty estimation, raising the upper-bound metric from 32%(F1) → 75%(Agreement).
- Built a retrieval-augmented pipeline with LLM-based similarity report feature + FAISS to inject neighborhood context for semantic deviation modeling, and fine-tuned Qwen3-4B (SFT + DPO) for a ~27% improvement over a zero-shot frontier model.
- Identified the “Consistency over Quantity” Paradox via category-wise analysis, showing that semantic alignment of comparison pairs drives performance more than raw dataset scale.

Learning Multi-Agent Collaborative Manipulation for Long-Horizon Quadrupedal Pushing

Pittsburgh, USA

Advised by Prof. Ding Zhao, CMU and Prof. Jie Tan, Google DeepMind [paper](#) [website](#) [github](#)

Feb. 2024 - Oct. 2024

- Developed a hierarchical multi-agent reinforcement learning framework for long-horizon planning and dynamic environment interactions, with a layered control architecture to enhance scalability and modularity.
- Designed and executed comprehensive benchmark experiments to evaluate system performance, achieving a 36% improvement in task success rates and a 24.5% reduction in execution latency compared to best baselines.
- Integrated the framework into a sim-to-real ML deployment pipeline, ensuring robust policy transferability through systematic feedback loops and real-world validation tests.

Collaborative Planar Pushing of Arbitrary Objects with Multiple Robots in Complex Scenes

Beijing, China

Advised by Prof. Meng Guo, Peking University [paper](#) [website](#)

Sept. 2023 - May. 2024

- Designed a hierarchical hybrid search algorithm that lowers execution time by 18% and improves trajectory smoothness by 67.8% compared to uniform path division.
- Proposed a fast and scalable multi-agent mode-switching method that theoretically ensures 100% collision-free operation, with switching path lengths guaranteed to be less than half the object’s circumference.
- Demonstrated the robustness of our planning method to motion and perception uncertainties, achieving a 100% success rate and 0.03 m average tracking error in high-fidelity simulations, and 0.05 m average tracking error in real-world hardware experiments.

TECHNICAL SKILLS

Functional: Machine Learning, SFT, DPO, RLAIF (PPO), RAG, LoRA, Agent, Prompt Engineering

Programming: Python, PyTorch, Jax, GitHub, L^AT_EX, Linux/Unix **Framework:** GCP, Docker, Isaac Gym, MuJoCo